UNIT I – Introduction to Android

1. What is Android?

Simple Definition

Android is an **Operating System (OS)** developed mainly by Google for smartphones, tablets, TVs, watches, cars, and many smart devices.

An Operating System manages both the hardware and software of a device.

Examples:

- Android Phone
- Android TV
- Smart Watch (Wear OS)
- Android Auto
- Tablet

Real-Life Analogy

Imagine buying a new car.

Without an engine, steering, gearbox, and control system, the car cannot run.

Similarly,

Your smartphone has

- Camera
- RAM
- Storage
- CPU
- Screen

But without Android OS, none of these components know how to work together.

Android acts as the **brain of the phone**.

Technical Definition

Android is an open-source, Linux-based mobile operating system that provides an application framework allowing developers to build applications using Java or Kotlin.

Real-Life Example

Suppose you click the Camera icon.

What happens?

Android

- Opens Camera App
- Connects Camera Hardware
- Starts Camera Driver
- Displays Preview
- Saves Image into Storage

All these operations are controlled by Android OS.

Industry Use Cases

Android is used in

- Smartphones
- Tablets
- Smart TVs
- Wearable Devices
- POS Machines
- Hospital Devices
- ATM Machines
- Smart Refrigerators

- Smart Cars
- Industrial Automation Devices

2. History of Android

Before Android

Around 2003

Most phones were

- Nokia
- Blackberry
- Windows Mobile
- Symbian

These phones had very limited applications.

No Play Store

No advanced touch interface

No AI

Limited internet support

Birth of Android

In 2003,

Four engineers started Android Inc.

Founders

- Andy Rubin
- Rich Miner
- Nick Sears
- Chris White

Their goal was to build software for smart mobile devices.

Real-Life Analogy

Think of Android Inc. as a small startup.

Initially,

Only a few people believed in smartphones.

Later,

Google saw its future and purchased Android.

Like buying a small startup that later became a global company.

Google Acquires Android

In 2005,

Google purchased Android Inc.

This completely changed the smartphone industry.

Google started developing Android as an Open Source Operating System.

First Android Phone

In 2008,

The first Android smartphone launched.

Device: HTC Dream (also called T-Mobile G1). HTC

Features

- Touch Screen
- Keyboard
- Gmail
- Google Maps
- Web Browser

- Android Market

Android Version Evolution

Version	Name	Major Feature
1.5	Cupcake	Virtual Keyboard
1.6	Donut	Better Search
2.0	Eclair	Google Maps Improvement
2.2	Froyo	Faster Performance
2.3	Gingerbread	Better Gaming
3.0	Honeycomb	Tablet Support
4.0	Ice Cream Sandwich	Modern UI
4.1–4.3	Jelly Bean	Smooth Performance
4.4	KitKat	Memory Optimization
5	Lollipop	Material Design
6	Marshmallow	Runtime Permissions
7	Nougat	Split Screen
8	Oreo	Picture-in-Picture
9	Pie	Gesture Navigation
10	Android 10	Dark Mode
11	Android 11	Better Privacy
12	Android 12	Material You
13	Android 13	Better Notification Control
14	Android 14	Security Improvements

Version	Name	Major Feature
15	Android 15	Enhanced Privacy and Performance

Why Are Android Versions Named?

Earlier versions were named after desserts in alphabetical order.

Example

Cupcake

↓

Donut

↓

Eclair

↓

Froyo

↓

Gingerbread

↓

Honeycomb

↓

Ice Cream Sandwich

↓

Jelly Bean

↓

KitKat

↓

Lollipop

↓

Marshmallow

↓

Nougat

↓

Oreo

↓

Pie

After Android 10,

Google stopped publicly using dessert names.

3. Android Operating System

What is an Operating System?

An Operating System manages

- CPU
- RAM
- Storage
- Display
- Camera
- Bluetooth
- WiFi
- Sensors

Real-Life Analogy

Imagine a school.

Teacher → Apps

Principal → Android OS

Students → Hardware

Without the principal,

everyone works independently and there is no coordination.

Android coordinates every hardware component.

Example

When WhatsApp opens,

Android

Checks RAM

↓

Allocates Memory

↓

Uses CPU

↓

Connects Internet

↓

Shows User Interface

↓

Stores Chat Database

Responsibilities of Android OS

- Memory Management
- File Management
- Process Management
- Device Security

- Network Management
- User Interface
- Hardware Control
- Power Management
- Permission Management

Technical Explanation

Android OS sits between the hardware and applications, providing APIs, system services, and security mechanisms so apps can safely access device resources.

Features of Android

- Open Source
- Multi-tasking
- Multi-touch Support
- Bluetooth
- Wi-Fi
- GPS
- Camera API
- Push Notifications
- Google Play Store
- Voice Assistant
- Widgets
- NFC
- Cloud Synchronization
- Security Permissions

Real-Life Example

Using Google Maps

Android uses

GPS

-

Internet

-

Storage

-

Display

-

CPU

All simultaneously.

4. Android Development Tools

Android applications are developed using specialized software.

1. Android Studio

Most popular IDE for Android Development.

Features

- Code Editor
- Emulator
- Debugger
- Layout Designer
- Profiler
- APK Builder

Real-Life Analogy

Android Studio is like a complete engineering workshop.

Instead of separately buying tools,

everything is available in one place.

Technical Example

Developer writes

Button btn = findViewById(R.id.button1);

Android Studio

- Highlights errors
- Suggests code
- Compiles program
- Creates APK

2. Java

Earlier,

Most Android applications were written using Java.

Example

System.out.println("Hello Android");

3. Kotlin

Modern Android applications mainly use Kotlin.

Google recommends Kotlin because it is concise, safer, and interoperates fully with Java.

Example

println("Hello Android")

4. Android SDK

SDK = Software Development Kit

It contains

- Libraries
- APIs
- Emulator
- Build Tools
- Platform Tools
- Documentation

Real-Life Analogy

Suppose you are cooking.

SDK is like a kitchen containing

- Stove
- Pan
- Spoon
- Knife
- Ingredients

Everything needed is already available.

5. Emulator

An Emulator behaves like a virtual Android phone on your computer.

Useful when a physical phone is unavailable.

Example

Developer

Writes App

↓

Runs Emulator

↓

Tests Application

↓

Fixes Bugs

6. ADB (Android Debug Bridge)

ADB connects the computer with an Android device.

Uses

- Install APK
- Debug App
- Copy Files
- Execute Commands
- View Logs

Example Command

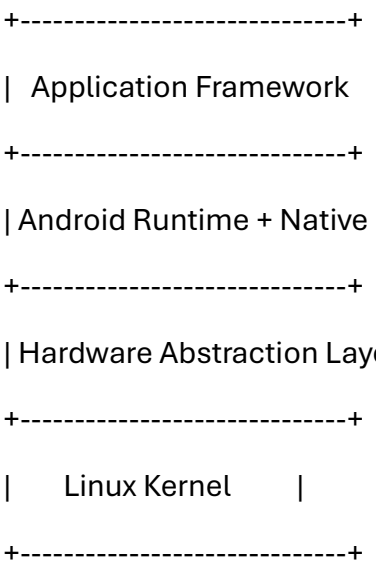
adb install app.apk

5. Android Architecture

Android architecture is organized into multiple layers, where each layer has a specific responsibility.

+------------------------------+

| Applications |



Real-Life Analogy

Think of a multi-story shopping mall.

Ground Floor → Security

First Floor → Building Services

Second Floor → Management

Third Floor → Shops

Customers only see the shops, but every floor works together to keep the mall running smoothly.

Similarly, Android applications rely on the lower layers to function.

Layer 1 – Linux Kernel

This is the foundation of Android.

Responsibilities:

- Memory Management
- Process Management
- Device Drivers

- Power Management
- Security
- Networking

Example: When you play music through earphones, the Linux Kernel communicates with the audio driver to send sound to the hardware.

Layer 2 – Hardware Abstraction Layer (HAL)

HAL acts as a bridge between hardware and Android software.

It provides standard interfaces for hardware components such as:

- Camera
- Bluetooth
- GPS
- Wi-Fi
- Fingerprint Sensor

Example: A camera app requests a photo through HAL, and HAL communicates with the specific camera hardware.

Layer 3 – Android Runtime (ART) & Native Libraries

Android Runtime (ART)

ART executes Android applications and manages memory efficiently.

Native Libraries

Provide ready-made functionality, including:

- SQLite (Database)
- OpenGL ES (Graphics)
- SSL (Security)
- Media Libraries (Audio/Video)
- Web Rendering

Example: A video streaming app uses media libraries to decode and play videos smoothly.

Layer 4 – Application Framework

This layer provides reusable services for app developers.

Common services include:

- Activity Manager
- Window Manager
- Package Manager
- Notification Manager
- Location Manager
- Resource Manager

Example: When a food delivery app sends an order update, the Notification Manager displays the notification on the phone.

Layer 5 – Applications

This is the top layer where users interact with apps.

Examples:

- Phone
- Camera
- Gmail
- Chrome
- WhatsApp
- Instagram
- Banking Apps

These apps use services from the lower layers rather than directly controlling hardware.

End-to-End Real-Life Flow

Suppose you take a selfie:

1. You tap the **Camera** app (Applications Layer).
2. The app requests camera services from the **Application Framework**.
3. **ART** executes the app, while native libraries process the image.
4. **HAL** communicates with the camera hardware.
5. The **Linux Kernel** controls the camera driver and storage.
6. The photo is captured and saved to your device.

Summary

Topic	**Key Point**
Android	Linux-based, open-source operating system for smart devices
History	Started by Android Inc. (2003), acquired by Google (2005), first phone released in 2008
Android OS	Manages hardware, software, memory, security, networking, and applications
Development Tools	Android Studio, Kotlin, Java, SDK, Emulator, ADB
Architecture	Five layers: Applications → Application Framework → Android Runtime & Native Libraries → HAL → Linux Kernel